

EXPERIMENT – 3

Performing Man-in-the-Middle (MITM) Attacks Using Ettercap in Kali Linux

Aim:

To perform and analyze a Man-in-the-Middle (MITM) attack using the Ettercap tool in Kali Linux for educational and ethical hacking purposes.

Software Requirements:

- **Host Operating System:** Windows / Linux / macOS
- **Virtualization Software:** Oracle VirtualBox
- **Guest Operating System:** Kali Linux
- **Security Tool:** Ettercap
- **System Requirements:**
 - RAM: Minimum 4 GB (8 GB recommended)
 - Disk Space: Minimum 30 GB
 - Processor: Intel/AMD with Virtualization support

Procedure:

Step 1: Start Kali Linux Virtual Machine

1. Launch Oracle VirtualBox.
2. Start the Kali Linux virtual machine.
3. Log in using Kali credentials.

Step 2: Verify Network Connectivity

1. Ensure Kali Linux is connected to the same network as the target system.
2. Check IP address using:
`ifconfig`

Step 3: Launch Ettercap

1. Open Terminal in Kali Linux.
2. Start Ettercap with graphical interface:
`ettercap -G`

Step 4: Select Network Interface

1. In Ettercap GUI, click Sniff → Unified Sniffing.
2. Select the active network interface (e.g., eth0 or wlan0).
3. Click OK.

Step 5: Scan for Hosts

1. Click Hosts → Scan for Hosts.
2. After scanning, select Hosts → Hosts List.
3. Identify the target system and gateway.

Step 6: Configure MITM Attack

1. Add the target system as Target 1.
2. Add the gateway/router as Target 2.

3. Click MITM → ARP Poisoning.
4. Enable Sniff remote connections.
5. Click OK.

Step 7: Start Sniffing

1. Click Start → Start Sniffing.
2. Observe intercepted traffic in real time.

Output:



```

Wireshark - Packet 886 - eth0
> Frame 886: Packet, 964 bytes on wire (7712 bits), 964 bytes captured (7712 bits) on interface eth0, id 0
> Ethernet II, Src: 52:55:0a:00:02:02 (52:55:0a:00:02:02), Dst: PCSSystemtec_63:b0:05 (08:00:27:63:b0:05)
> Internet Protocol Version 4, Src: 142.250.77.131, Dst: 10.0.2.15
> Transmission Control Protocol, Src Port: 80, Dst Port: 51724, Seq: 1, Ack: 430, Len: 910
> Hypertext Transfer Protocol
> Online Certificate Status Protocol

0000 08 00 27 63 b0 05 52 55 0a 00 02 02 08 00 45 08 ...c..RU...E
0010 03 b6 f9 9d 00 00 40 06 95 10 8e fa 4d 83 0a 00 .....@...M...
0020 02 0f 00 50 ca 0c 01 59 c6 8c 5b 23 40 32 50 18 ...P...Y...[#@2P
0030 ff ff 2c 31 00 00 48 54 54 50 2f 31 2e 31 20 32 ...1..HT TP/1.1.2
0040 30 30 20 4f 4b 0d 0a 43 6f 6e 74 65 6e 74 2d 54 00 OK..Content-T
0050 79 70 65 3a 20 61 70 70 6c 69 63 61 74 69 6f 6e ype: app lication
0060 2f 6f 63 73 70 2d 72 65 73 70 6f 6e 73 65 0d 0a /ocsp-re sponse..
0070 44 61 74 65 3a 20 4d 6f 6e 2c 20 30 39 20 4d 61 Date: Mo n, 09 Ma
0080 72 20 32 30 32 36 20 30 39 3a 34 37 3a 31 34 20 r 2026 0 9:47:14
0090 47 4d 54 0d 0a 43 61 63 68 65 2d 43 6f 6e 74 72 GMT: Cac he-Contr
00a0 6f 6c 3a 20 70 75 62 6c 69 63 2c 20 6d 61 78 2d ol: publ ic, max-
00b0 61 67 65 3d 31 34 34 30 30 0d 0a 43 6f 6e 74 65 age=1440 0 Conte
00c0 6e 74 2d 53 65 63 75 72 69 74 79 2d 50 6f 6c 69 nt-Secur ity-Poli
00d0 63 79 2d 52 65 70 6f 72 74 2d 4f 6e 6c 79 3a 20 cy-Repor t-Only:
00e0 73 63 72 69 70 74 2d 73 72 63 20 27 6e 6f 6e 65 script-s rc 'none
00f0 27 3b 20 66 6f 72 6d 2d 61 63 74 69 6f 6e 20 27 '; form- action '
0100 6e 6f 6e 65 27 3b 20 66 72 61 6d 65 2d 73 72 63 none'; f rame-src

```

No.	Time	Source	Destination	Protocol	Length	Info
239	13.238583915	10.0.2.15	142.250.77.131	OCSP	482	[TCP Previous segment not captured] Request
243	13.275598126	142.250.77.131	10.0.2.15	OCSP	966	Response
878	15.782195708	10.0.2.15	142.250.77.131	OCSP	482	[TCP Previous segment not captured] Request
884	15.816021076	10.0.2.15	142.250.77.131	OCSP	482	[TCP Previous segment not captured] Request
886	15.827514745	142.250.77.131	10.0.2.15	OCSP	964	Response
897	15.855424016	142.250.77.131	10.0.2.15	OCSP	964	Response
1629	28.962820845	10.0.2.15	142.250.77.131	OCSP	481	Request
1631	28.962883669	10.0.2.15	142.250.77.131	OCSP	481	Request
1686	29.003374444	142.250.77.131	10.0.2.15	OCSP	1156	Response
1692	29.006338296	142.250.77.131	10.0.2.15	OCSP	964	Response
1746	29.046240399	10.0.2.15	142.250.77.131	OCSP	481	Request
1779	29.086900884	142.250.77.131	10.0.2.15	OCSP	964	Response

```

> Frame 239: Packet, 482 bytes on wire (3856 bits), 482 bytes captured (3856 bits) on interface eth0, id 0
> Ethernet II, Src: PCSSystemtec_63:b0:05 (08:00:27:63:b0:05), Dst: 52:55:0a:00:02:02 (52:55:0a:00:02:02)
> Internet Protocol Version 4, Src: 10.0.2.15, Dst: 142.250.77.131
> Transmission Control Protocol, Src Port: 51698, Dst Port: 80, Seq: 2, Ack: 1, Len: 428
> Hypertext Transfer Protocol
> Online Certificate Status Protocol

0000 52 55 0a 00 02 0000 01 d4 2e 2e 40
0010 01 d4 2e 2e 40 0020 4d 83 c9 f2 00
0030 e8 aa ea 52 00 0040 48 54 54 50 2f
0050 6f 2e 70 6b 69

```

Result:

The MITM attack was successfully performed using Ettercap in Kali Linux. The experiment demonstrated how insecure network communication can be intercepted, highlighting the importance of encryption and secure network protocols.